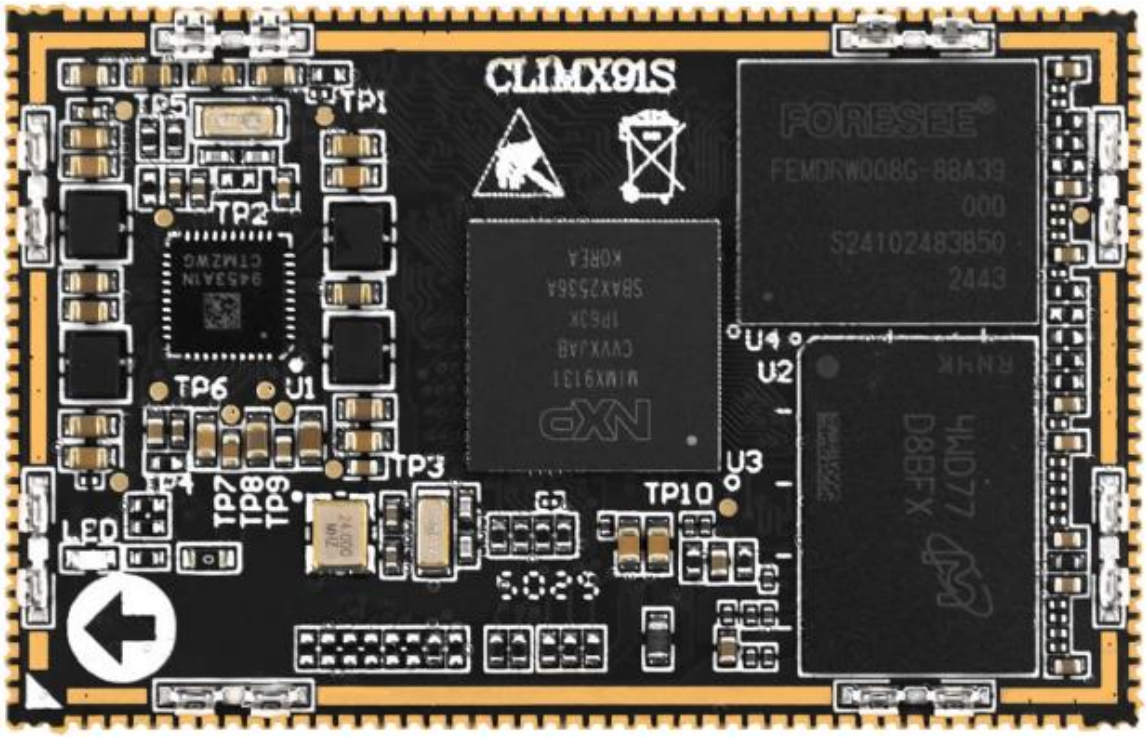
ATK-CLIMX91S

Core Board Specification

V1.0



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Revision History:

Version	Version Update Notes	Responsible person	Proofreading	Date
V1.0	release officially	ALIENTEK Linux Team	ALIENTEK Linux Team	2026.04.01

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Brief

The ATK-CLIMX91S core board is a high-performance embedded development platform developed by ALIENTEK based on the NXP i.MX9131 chip, specially designed for embedded system development. This specification document elaborates on the technical specifications, functional features and application scenarios of the core board in detail. It aims to provide comprehensive reference materials for developers and engineers, enabling them to fully leverage the advantages of this core board in product design and development.

Chapter 1. Core board overview

1.1 Core board introduction

ATK-CLIMX91S is a low-power embedded Linux development platform launched by ALIENTEK, based on the NXP i.MX 91 series processors. It is specially designed for application scenarios such as smart home controllers, networked home appliances, home entertainment systems, industrial equipment and medical platforms.

Equipped with the i.MX 91 processor, the platform integrates one 64-bit RISC Arm Cortex-A55 core with a maximum main frequency of up to 1.4 GHz. It supports LPDDR4 memory with a bandwidth up to 2.4 GT/s and integrates inline ECC, which effectively improves system stability and data integrity. Meanwhile, it is configured with 8 GB eMMC storage to meet the demand for large-capacity data storage.

In terms of interface resources, ATK-CLIMX91S (corrected from the original ATK-DLIMX91S) provides rich peripheral support, including dual USB 2.0 Type-C ports, dual Gigabit Ethernet ports, dual CAN-FD interfaces, 8-channel UARTs, and one 4-channel 12-bit ADC. It can meet the connection requirements of various peripherals and is highly suitable for complex application environments such as industrial real-time communication.

ALIENTEK provides comprehensive software and documentation resources for the ATK-CLIMX91S core board and its matching baseboard, including U-Boot, Linux kernel, peripheral driver source code, file system, Qt source code, and C application source code, as well as relevant development tools and development environments.

The documentation includes detailed tutorials and user manuals, covering guides for Linux embedded driver development, Linux C application programming, and more. In addition, complete hardware materials are provided, such as the PDF schematic diagram of the baseboard, AD component libraries, mechanical dimension drawings of the development board, and the i.MX91 processor reference manual.

These abundant software and hardware resources help developers fully utilize the functions and performance of the core board, and accelerate the development and optimization of embedded systems.

1.2 Application Fields

Figure 1.2-1 Partial Application Fields

1.3 Document Download

Download Center：http://www.openedv.com/docs/boards/arm-linux/iMX91hxb.html

Chapter 2. Product Selection

2.1 Product Naming

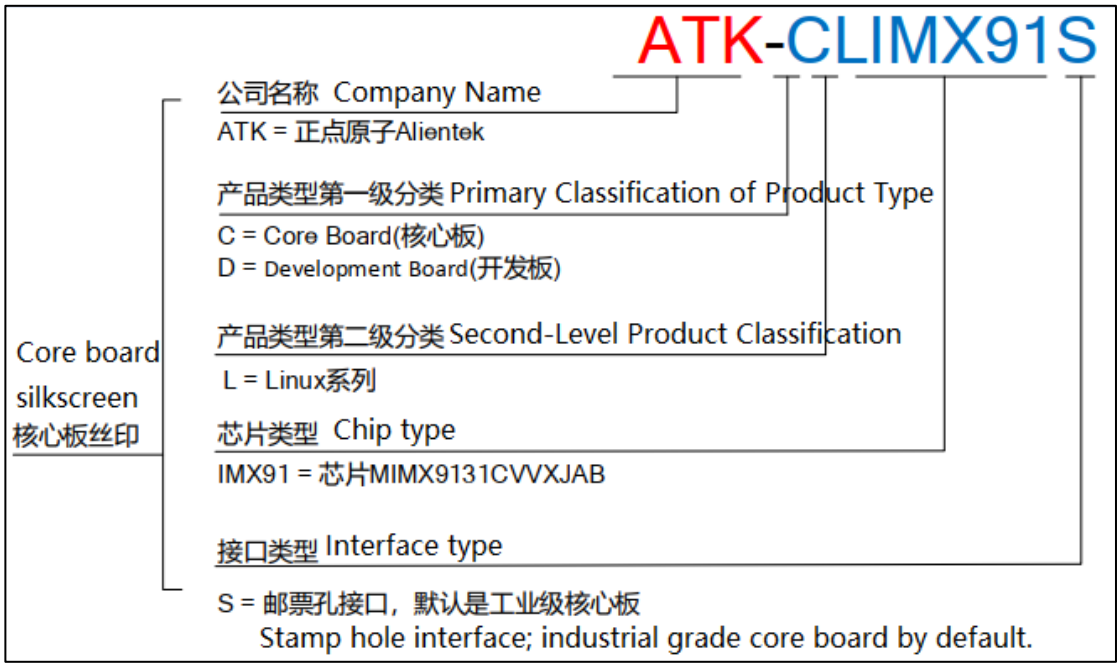

Figure 2.1-1 Product Naming Rules

2.2 Description of Industrial Grade

The ATK-CLIMX91S series core boards are only available in industrial grade versions.
The operating temperature range of the industrial-grade core board is -40℃ ~ +85℃.
The silkscreen marking of the industrial-grade stamp hole version core board is CLIMX93S.
Unless otherwise specified, all pictures in this document are of the industrial-grade core board.

Chapter 3. Electrical Characteristics

3.1 Hardware Parameters

Item	Parameter	Remark
Core Board Dimension Specifications	50mm*32mm	
Processor Model	NXP i.MX9131	LFBGA306 package
Processor Architecture	Single core 1.4GHz Cortex-A55	
Power Management	Integrated Power Supply	
Memory	1GB LPDDR4 (standard configuration), with customized capacities available for mass production.	The actual silkscreen label after mounting shall prevail.
Storage	8GB eMMC (standard configuration), with customized capacities available for mass production.	The actual silkscreen label after mounting shall prevail.
Operating Voltage	5V1A	Supply Voltage Range：5V±200mV
Power Consumption	＜1W	minimum system power consumption
Operating Temperature	Industrial Grade:-40℃~ +85℃	
Interface Pin Count	150PIN	
Pin Pitch	1mm	
Interface Type	Stamp hole	
PCB Process	8-layer, independent ground signal layers, immersion gold，lead-free process	

3.2 Pin Order and Interface Signals

The pin definitions corresponding to the stamp hole core board are provided in the PINOUT section of the ATK-CLIMX91S core board schematic, with a total of 150 pins.

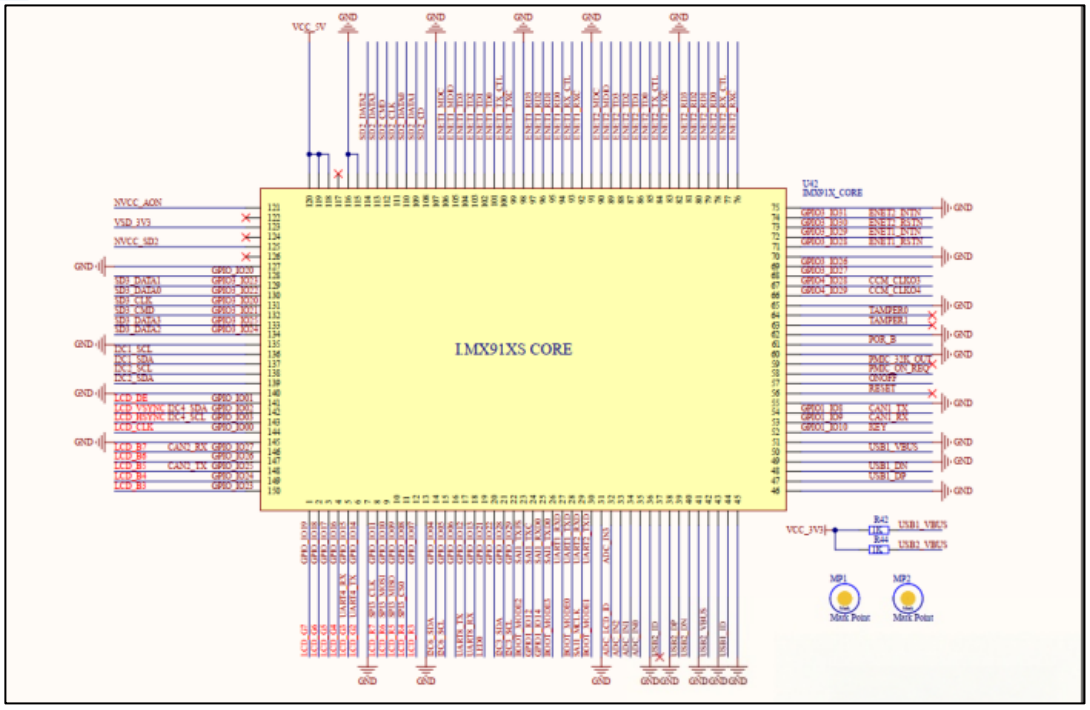

Figure 3.2-1The pin definition section of the baseboard schematic

3.3 Pin Multiplexing Function Resources

The core board leads out all I/O signals of the processor. Users can design customized baseboards as required to utilize the on-board I/O resources and configure the pins to alternative multiplexing functions.

In accordance with peripheral functions, this section lists the maximum available quantity of each multiplexable peripheral on the ATK-CLIMX91S core board. For detailed selection, please refer to the chip datasheet.

(Maximum quantity of a single peripheral: the maximum number of a specific peripheral that can be used on the core board without occupying other peripheral resources.)

Peripheral Function	Maximum Multiplexing Quantity of Single Peripheral	Peripheral Function	Maximum Multiplexing Quantity of Single Peripheral
GPIO	128	uSDHC	3, SD/SDIO/eMMC
USB	2	RGB LCD	1, 24 bit
Ethernet	2	Parallel CSI	1, 8 bit, YUV/RGB
CAN-FD	2	SAI	3
LPUART	8	Timer/PWM	6
LPI2C	8	Timer	2
LPSPI	8	WDOG	5
I3C	2	JTAG	1
I2C	4	FlexIO	2
ADC	1, 4 Channels, 12 bit		

Chapter 4. Product Resources

4.1 Core Board Appearance

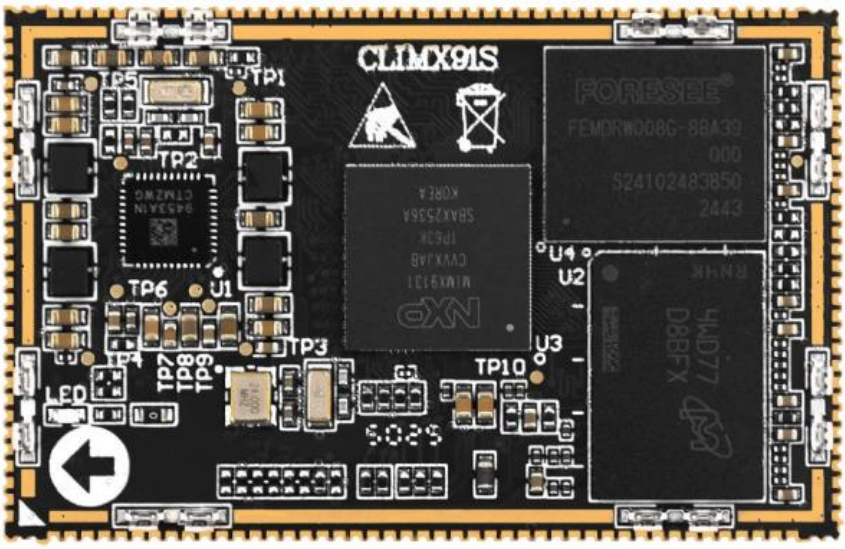

Figure 4.1-1 Core Board Front

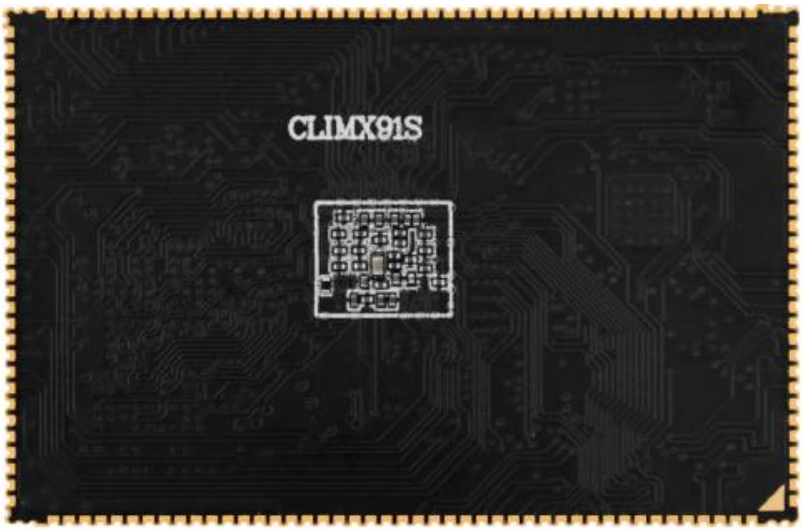

Figure 4.1-2 Core Board Back

4.2 Core Board Hardware Resource Distribution Diagram

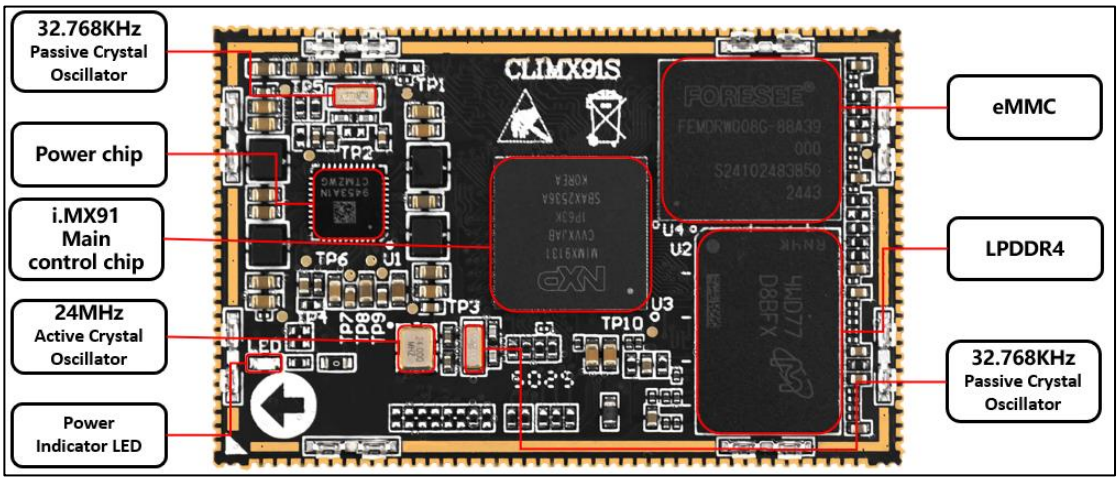

Figure 4.2-1 Core Board Hardware Resource Distribution Diagram

4.3 Core Board Mechanical Dimension Drawing

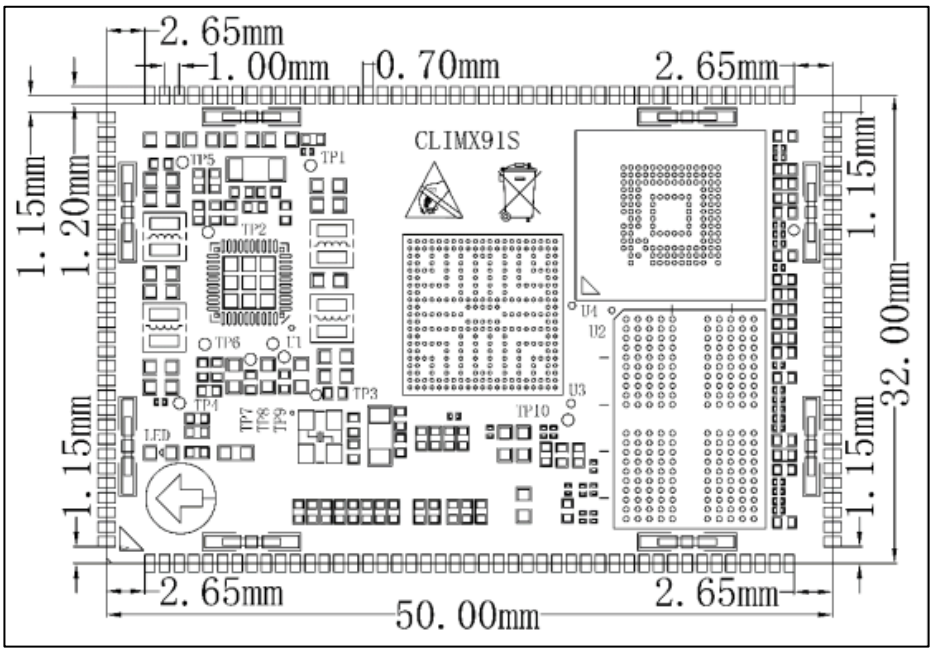

Figure 4.3-1 Mechanical Dimension Drawing of Core Board Front Side

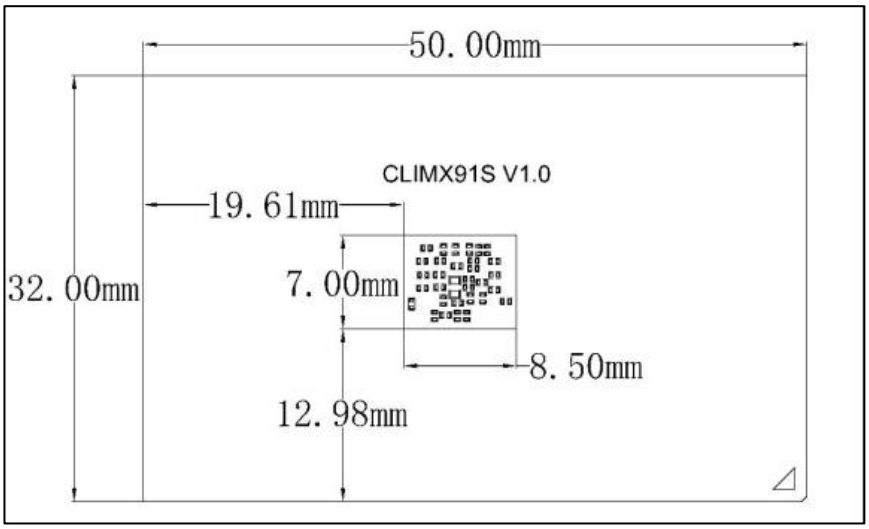

Figure 4.3-2 Core Board Rear Mechanical Dimension Drawing

4.4 Core Board Software Resources

4.4.1 Software Resources Summary

Type	Description	Remark
U-Boot	Version:2024.04	Source Code Provided
Linux kernel	Version:6.6.52	Source Code Provided
Yocto Root File System	Yocto Root File System	Image File Provided
Qt	Version:5.15	Source Code Provided
Cross Compiler	Fsl-imx-xwayland, version:13.3.0	Software Provided
System Flashing	UUU tool and SD card	Tutorials Provided

4.4.2 Driver Support Table

Driver Type	Corresponding Module / Node	Status	Remarks
UART	lpuart1、lpuart8	Supported	LPUART1 as the console
GPIO key	gpio_keys	Supported	User Keys
GPIO LED	leds	Supported	System Indicators
I2C Bus	lpi2cl, lpi2c2, lpi2c6	Supported	Three-channel I2C Enabled
EEPROM	eeprom@50（AT24C64）	Supported	I2C EEPROM
PMIC/ Power	pmic@32（PF9453） + fixed regulators	Supported	Including BUCK / LDO / Fixed Power Supplies
Audio CPU DAI	sail	Supported	MCLK output configured
Audio Codec	es8390@11	Supported	I2C Audio Codec
Sound Card Binding	Simple-audio-card	Supported	I2S Sound Card Link
ADC	adc1	Supported	External vref_1v8

CAN	flexcan1	Supported	Drivers are built in module mode
Mailbox/MU	mu1、mu2	Supported	Message Unit
RPMSG	MU related channels	Supported	Partial components adopt module mode
Gigabit Ethernet Port1	eqos	Supported	RGMII interface
Gigabit Ethernet Port2	fec	Supported	RGMII interface
PHY Driver	ethphy1、ethphy2	Supported	Realtek PHY
USB OTG	usbotg1	Supported	OTG mode
USB Host	usbotg2	Supported	Host mode
eMMC	usdhc1	Supported	Onboard Storage
SD card	usdhc2	Supported	4-bit, card detect
Watchdog	wdog3	Supported	system watchdog
Temperature Sensor	tempsense0	Supported	SoC temperature detection
Touch Screen	goodix_ts@14(GT9271/GT911)	Supported	Driver in module mode
RGB display	lcdif+parallel_disp_fmt+panel-dpi	Supported	RGB parallel display
Backlight	panel-rgb-backlight	Supported	GPIO Backlight Control

4.5 Core Board Adapted Modules

At present, the core board has completed the adaptation of several functional modules when matched with the baseboard developed by our company. For more information on additional adapted modules supported by the ATK-DLIMX91S development board, please contact the official technical support of ALIENTEK.

Note: When testing the following modules with the ATK-DLIMX91S development board, a dedicated RGB LCD adapter test board is required.

The PCB design files of this test board have been open-sourced and released in the cloud storage directory of the corresponding development board materials:

Development Board Disc A – Basic Materials\06. Hardware Documents\06. RGB LCD Display Adapter Test Board.

Users may download the files independently and complete PCB fabrication and verification according to actual project requirements.

正点原子旗舰店
4.3寸RGBLCD触摸屏模块
RGB屏 | 电容屏 | 800*480
支持5点同时触摸
提供ST驱动

正点原子旗舰店
7寸RGBLCD触摸屏模块800
RGB屏 | 电容屏 | 800*480
支持5点同时触摸
提供ST驱动

正点原子
7寸RGBLCD触摸屏模块1024
RGB屏 | IPS电容屏 | 1024*600
支持5点同时触摸
提供ST驱动

正点原子旗舰店
10.1寸RGBLCD触摸屏模块
RGB屏 | 电容屏 | 1280*800
支持10点同时触摸
RGB/LVDS

Chapter 5. Precautions and maintenance

5.1 Notes

- Do not plug and unplug peripheral modules with power!
- Before using the product, please carefully read this manual and related development manuals, and pay attention to the applicable matters of the platform.
- Follow all instructions and warnings on the product.
- Please use this product in a cool, dry and clean place.
- Please keep the product dry. If any liquid splashes or soaks, power off immediately and let dry thoroughly.
- Do not use organic solvents or corrosive liquids to clean the product.
- Do not use or store this product in dusty, dirty and messy environment.
- If not used for a long time, please package this product, pay attention to moisture-proof and dust-proof.
- Pay attention to the ventilation and heat dissipation of the product during use to avoid component damage caused by excessive temperature during operation.
- Do not use this product in alternating hot and cold environment to avoid dew damage to components.
- Do not treat this product roughly, drop, knock or shake violently may damage the line and components.
- Pay attention to anti-static when using this product.
- FPC flexible cable is fragile, when plugging cable, pay attention to check whether the metal at both ends of the cable is misplaced and falling off.
- All products have passed the product test before shipment. Please use the development board corresponding to the ALIENTEK for power on test for the first time.
- Do not repair or disassemble the company's products by yourself. If the product fails, please contact the company in time for maintenance.
- Unauthorized modification or use of unauthorized parts may damage the product, the resulting damage will not be repaired.

5.2 After-sales Support

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Taobao shop: ALIENTEK flagship store
Business Contact: **ALIENTEK Flagship Store**, transfer to Business Customer Service
Forum: http://www.openedv.com/forum.php